Creativity and Innovation

Chapter 2: Processes (Steps) of Creativity and Innovation

Day 2 (Thursday) Morning and Afternoon Sessions

From explanation to action

Yesterday we studied theories that explain where creativity comes from.

Today we ask a practical question. How does a vague problem travel to a working solution that helps real people?

Both creativity and innovation follow recognizable steps. When you know the steps, you stop waiting for inspiration and start managing it.

Plan for the day

Block	Topic	Min
Morning	Welcome, Day 1 recap, 2.1 Introduction to the creativity process	15
Morning	2.2.1 Preparation 2.2.2 Incubation	20 + 20
Morning	Break	10
Morning	2.2.3 Illumination 2.2.4 Evaluation 2.2.5 Implementation and recap	15 each
Afternoon	Welcome back, review, 2.3 Introduction to the innovation process	15
Afternoon	2.3.1 Ideation and 2.3.2 Concept Development	30
Afternoon	2.3.3 Prototyping	15
Afternoon	Break	10
Afternoon	2.3.4 Validation and 2.3.5 Deployment and Scaling	25
Afternoon	2.4 Creativity and Innovation together, two-minute pitches	25

Learning outcomes

By the end of Day 2, you will be able to:

1. Describe the five steps of the creativity process and the five steps of the innovation process.
2. Apply practical tools at each step: problem questions, incubation breaks, scoring matrices, SCAMPER, prototypes, and test plans.
3. Explain how creativity and innovation depend on each other and how they differ.
4. Identify ways organizations can strengthen both, and the challenges that block them.
5. Develop one real workplace idea from problem to rollout plan and present it in two minutes.

How we will work today

Key ideas

A short summary of each step.

Activity

Group work, done on your feet.

Project thread

Your own project. Keep every answer, because each step uses the work of the previous one.

Check your understanding

Questions to test yourself.

One problem travels with you all day

In the first hour, each group chooses one real problem from its workplace.

You improve it step by step. By the end of the day you leave with a tested idea and a plan, not only notes.

To see each step in action, we follow **Hana**, who works in a university student records office. Students wait about three hours for a clearance letter.

Morning Session

The five steps of the creativity process

2.1 What is a process?

A process is a series of steps that lead to a result. We follow one when we cook injera, register a student, or process a leave request.

Creativity can seem like a gift that arrives without warning. Yet studies of inventors, writers, and scientists show recognizable stages.

Graham Wallas (1926) proposed four stages: preparation, incubation, illumination, verification.

The five steps we will use

We keep Wallas's first three and replace verification with two practical steps: **Evaluation** (is this idea good enough?) and **Implementation** (how do we put it to use?).

Real creative work is not a straight road. People often jump back, for example from Evaluation to Preparation when an idea fails. Treat the steps as a map, not a strict timetable.

2.2.1 Preparation: understand before you solve

Preparation is where you understand the problem and learn what is already known.

Many people skip it and jump to solutions. That is a mistake. An answer to the wrong question has no value, however clever it is.

It includes defining the problem, collecting facts, observing how things really work, talking to the people affected, and learning from earlier attempts.

Preparation: useful tools

How might we Write the problem as an open question that invites many answers.

The five whys Ask “why” five times to move from a symptom to a root cause.

Observation and interviews Watch the process and ask users what is hard about it.

Facts and numbers Count, measure, and map the steps before you change them.

Preparation: Hana's example

Hana in the records office

Hana counts how many students arrive each hour and how long each request takes. She asks students why they wait.

The five whys show that most students come on the last days of the semester, because nobody told them that clearance letters can be requested earlier.

Her question

"How might we reduce the time students wait for a clearance letter from three hours to thirty minutes?"

Preparation: key ideas

Key ideas

- Preparation means understanding the problem before looking for solutions.
- A well-written “How might we” question guides the whole process.
- Facts, observation, and conversation with users are the raw material of later ideas.

Project thread 1: choose and frame your problem

10 minutes Groups of three or four

Choose one real problem from your workplace that you can influence: long waiting times, lost documents, repeated errors, confusing instructions. Then write:

- Your problem as a “How might we” question, with a target if possible.
- Three facts you still need, and how you could find them.
- Two people you would ask: a client, a colleague, or a supervisor.

Check your understanding: Preparation

Discuss with a partner

1. Why is it risky to begin generating solutions before the problem is understood?
2. Turn this complaint into a "How might we" question: *"Our files are always lost."*

2.2.2 Incubation: step back on purpose

After intense preparation you may feel stuck. The next step is surprising: stop working on the problem for a while.

The mind keeps working in the background. Many people report solutions arriving in the shower, on a walk, in a taxi, or after a night of sleep.

Research reviews suggest a short break on an undemanding activity can help with open-ended problems.

Why a break can work

- We stop repeating the same unsuccessful approach.
- Weaker memories and loose connections get a chance to surface.
- We return with a fresh view.

Not laziness, not procrastination

Incubation works only after real effort in Preparation, and it needs a time limit.

Ways to incubate

- Take a short walk or do light physical activity.
- Switch to a different, easy task for a short time.
- Sleep on it, and keep paper and pen near your bed.
- Talk about something unrelated, or talk to someone from a different field.
- Return at a planned time and note what new thoughts appeared.

Hana in the records office

After two days of data and interviews, Hana is tired. She helps with her cousin's wedding preparations and sees guests receive numbered seats. They wait calmly because they know when their turn comes.

Incubation: key ideas

Key ideas

- Incubation is a planned pause that lets the mind work in the background.
- It follows serious preparation and does not replace it.
- Light activity, rest, and unrelated conversation are good incubators.

Activity 1: a planned pause

10 minutes

1. Write your project question at the top of a page.
2. Stand up and take a five-minute silent walk. Do not discuss the problem.
3. Return and write down any new thoughts, even if they seem silly.
4. List three incubation habits you could use at work.

Check your understanding: Incubation

Discuss with a partner

1. Explain the difference between incubation and procrastination.
2. Which incubation activity works best for you, and why?

2.2.3 Illumination: the “aha” moment

Illumination is the moment a new idea or connection suddenly appears. It feels like a light switching on.

It is the most dramatic step and the most misunderstood. The flash does not come from nothing. It is the visible tip of the work done in Preparation and Incubation.

Insights are small and fragile. They can vanish within minutes. Often the first insight is only a piece of the solution, which we shape step by step, as the Honing Theory from Day 1 described.

How to welcome an insight

1. Keep a notebook or phone note always available.
2. Write the idea at once in a few words, with a quick sketch.
3. Do not judge the idea yet. Evaluation comes next.
4. Build on it. Ask what else could be combined with it.

Hana in the records office

On her way to work, Hana sees patients holding appointment cards at a health centre. She realizes students could book a time slot for their letters, and most requests could be prepared in advance. She writes it down at once.

By evening she has two more ideas: an information board telling students when to apply, and a simple online request form.

Illumination: key ideas

Key ideas

- Illumination is the sudden appearance of an idea after preparation and a pause.
- Insights are fragile, so capture them immediately.
- Do not judge ideas at this moment. Collect first, judge later.

Project thread 2: idea burst

5 minutes Work alone

Look again at your “How might we” question. Write as many ideas as you can in five minutes.

Quantity matters more than quality right now. Then mark your three favourites with a star.

Check your understanding: Illumination

Discuss with a partner

1. Why should we avoid criticizing an idea at the moment it appears?
2. What was the source of Hana's insight in the example?

2.2.4 Evaluation: not every idea deserves action

Evaluation tests ideas against clear criteria and decides which are worth developing.

It connects to the **Evolution Theory** from Day 1, where selection decides which ideas survive. It also connects to **Systems Theory**: the people who accept or reject the idea (the field) must be considered.

Common criteria: useful, new, feasible, affordable, accepted.

A scoring matrix: 1 (weak) to 5 (strong)

Idea	Useful	New	Feasible	Affordable	Accepted	Total
A. Appointment slots for letters	5	3	4	4	4	**20**
B. Open two extra counters	3	1	3	2	4	13
C. Online request form	5	4	3	3	3	18

The numbers support a conversation. They do not make the decision.

Hana's group chooses A. C can follow later, since the university does not yet have the system. A low score on newness did not stop A, because usefulness and feasibility mattered more here.

Evaluation: key ideas

Key ideas

- Test ideas against clear criteria: useful, new, feasible, affordable, accepted.
- A scoring matrix supports discussion and does not replace judgment.
- Consider the people who must accept the idea.

Project thread 3: score your ideas

10 minutes

1. Take your three starred ideas.
2. Score each from 1 to 5 on the five criteria and add the totals.
3. Choose the idea you will develop.
4. Write one sentence explaining why you chose it.

Check your understanding: Evaluation

Discuss with a partner

1. Name three criteria you can use to evaluate an idea.
2. Why should the final decision not rest on scores alone?

2.2.5 Implementation: start small and act

An idea that stays in a notebook has produced no change.

At this stage, implementation means starting small: prepare a clear action plan, tell the people involved, and try the idea in a limited way.

This grows into the formal innovation process that we study this afternoon.

The action plan: five questions

1. What exactly will we do first?
2. Who is responsible for each action?
3. When will each action be done?
4. What resources do we need?
5. How will we know it is working?

Expect resistance

People worry about extra work, about losing control, or about failure.

Explain the benefits in simple words, listen to concerns, and invite colleagues to help improve the idea.

Remember

An idea that people help to shape is an idea that people help to defend.

Hana in the records office

Hana asks her supervisor for permission to try appointment slots for one week with fifty students. She prepares a simple sign-up sheet and announces it on the notice board.

Project thread 4: action plan

10 minutes

Fill in the five-question table for your chosen idea. Be specific about who does what by when.

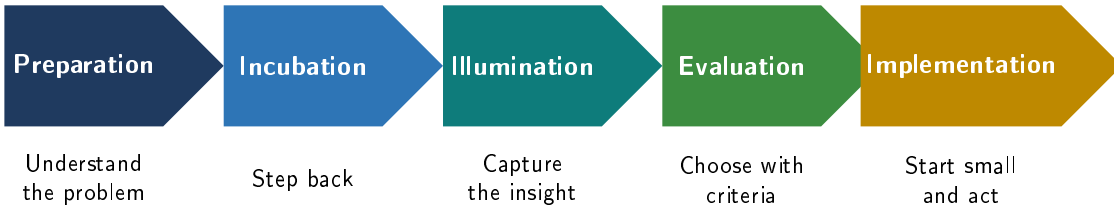
Share your plan with another group and ask them to find **one weakness**.

Check your understanding: Implementation

Discuss with a partner

1. Why is it wise to start the implementation in a small and limited way?
2. How can you reduce resistance from colleagues?

Morning recap



Before lunch

Write the name of the person in your office whom you will ask for feedback on your project.

Afternoon Session

The five steps of the innovation process
and how creativity and innovation work together

Welcome back: quick review

5 minutes No notes

Write the five steps of the creativity process in the correct order and give one tool for each. Compare with a partner.

2.3 From idea to value that people use

This morning we looked at how a person or small team generates and shapes an idea.

Innovation goes further. It is the full journey from idea to an improvement that real people use and value. It is more organized, involves more people, needs resources, and depends on evidence.

The output of the creativity process becomes the input of the innovation process.

The five steps of innovation

Names differ between organizations, but the logic is the same. These steps appear in design thinking and in product and service development.

Hana's tested idea of appointment slots now enters a more formal journey.

2.3.1 Ideation: widen, then narrow

Ideation is the deliberate, structured generation of ideas, usually in a group. The team plans it, uses tools, and builds on one another's thoughts.

Two phases: first **widen** (many ideas), then **narrow** (the strongest).

Alex Osborn, who invented brainstorming, gave four rules:

- Delay judgment.
- Welcome wild ideas.
- Aim for quantity.
- Build on the ideas of others.

SCAMPER: seven questions that change what exists

	Question	Example for the records office
S	Substitute: what can we replace?	Replace paper request forms with a phone message
C	Combine: what can we join together?	Combine clearance and transcript requests in one visit
A	Adapt: what can we borrow from elsewhere?	Adapt the appointment cards of health centres
M	Modify: what can we enlarge, reduce, or change?	Shorten the form to one page
P	Put to another use: how else can it be used?	Use the waiting area as a help desk
E	Eliminate: what can we remove?	Remove the second signature for simple requests
R	Reverse: what if we turn it around?	Staff go to departments, not students to the office

Project thread 5: SCAMPER round

10 minutes

Apply the seven SCAMPER questions to your chosen idea and write at least one answer for each.

Then select the one or two changes that would make your idea stronger.

2.3.2 Concept development: one page, simple words

A promising idea is still only a sentence. Concept development turns it into a clear description that others can understand, discuss, and approve.

A good concept answers **who, what, why, and how**. Writing it down on one page forces the team to find gaps.

The concept sheet

1. Who is this for?
2. What problem does it solve?
3. How does it work, in three steps?
4. What benefits will it bring?
5. What resources and approvals are needed?
6. What are the main risks?

Hana in the records office

For students who need clearance letters, the office offers thirty-minute time slots. Students sign up by phone or at the notice board, staff prepare letters before the slot, and students collect them within five minutes.

Benefits: shorter waiting, calmer staff. Needs: a shared schedule and the registrar's approval. Main risk: students who miss their slot.

Project thread 6: write your concept sheet

10 minutes

Complete the six-question concept sheet for your chosen idea.

Keep each answer to one or two lines.

2.3.3 Prototyping: learn cheaply

A prototype is a quick, cheap, rough version of the idea that lets people see it, touch it, or try it.

Its purpose is to **learn, not to impress**. It is far cheaper to find a mistake on paper than after the whole system is built. This is called failing early and cheaply.

Kinds of prototypes

- A **sketch or storyboard** that draws the steps a user follows.
- A **paper mock-up** of a form, sign, or screen.
- A **role play** in which colleagues act out the new process.
- A **minimum viable version**: the simplest working version that tests the main idea with real users.

Hana in the records office

Hana does not wait for a computer system. She prints a sign-up sheet with time slots, places it at the office door, and offers the service for one week to fifty volunteer students. The tools cost almost nothing.

Activity 2: five-minute prototype

10 minutes

Draw a four-box storyboard of your idea. Stick figures are welcome.

1. The user with the problem.
2. The user meeting your solution.
3. How the solution works.
4. The result.

Show it to a neighbor and ask what is unclear.

2.3.4 Validation: does it really work?

Validation tests the prototype with real users, collects evidence, and compares results with your goals.

Do not rely only on your own opinion or on friendly colleagues who want to please you.

Observe what users do, not only what they say.

A test plan: Hana's example

Test plan question	Hana's answer
What do we want to learn?	Do time slots reduce waiting, and do students use them?
Who will test it, and for how long?	Fifty students for one week
What will we measure?	Average waiting time, missed slots, student comments
What result means success?	Waiting below thirty minutes and fewer than one in ten missed slots

After the trial: three decisions

1. **Continue** with the idea.
2. **Adjust** it and test again.
3. **Stop** and try a different idea.

Stopping is not failure. It is a wise use of resources when the evidence is clear.

Hana in the records office

Waiting fell to twenty minutes, but some students missed their slot. Hana adds a reminder message and continues.

Project thread 7: test plan

5 minutes

Fill in a four-row test plan like Hana's for your own idea, with a clear success measure.

2.3.5 Deployment and scaling

When the evidence is good, the idea is **deployed**: rolled out for regular use. **Scaling** extends it to more people, offices, or locations.

This is where many good ideas fail. The excitement of the pilot ends and the hard daily work begins. A careful rollout reduces the risk.

A careful rollout

1. **Name an owner** who is responsible for the new practice.
2. **Train** the people who will use it and give simple written guidance.
3. **Communicate** why the change is happening and what each person must do.
4. **Roll out in phases**: one office, a few, then all.
5. **Support and monitor**: collect feedback, fix problems quickly, track results.
6. **Make it standard**: update procedures, budgets, and job descriptions.

Care when scaling

An idea that worked in one office may need changes in another.

The **Shrinking Earth Theory** from Day 1 reminds us to adapt, not only to copy.

Hana in the records office

In the second month the office extends the slots to the transcript service. Then Hana shares the method with other university offices.

Innovation process: key ideas

Key ideas

- Ideation widens and then narrows ideas, with tools such as brainstorming rules and SCAMPER.
- A concept sheet describes the idea on one page. A prototype lets people try it cheaply.
- Validation tests with real users and ends in a decision: continue, adjust, or stop.
- Deployment and scaling need an owner, training, communication, phased rollout, and monitoring.

Check your understanding: Innovation

Discuss with a partner

1. Why should a prototype be quick and rough?
2. What three decisions can follow a validation test?
3. Name three things that help an idea survive the scaling step.

2.4 Creativity and Innovation

Two processes, one system

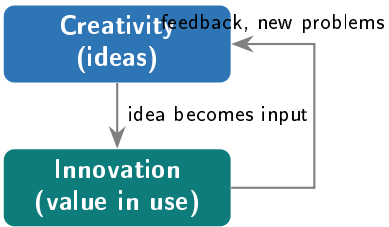
2.4.1 They need each other

Creativity without innovation produces many ideas that nobody uses.

Innovation without creativity repeats old solutions with new labels.

Creativity supplies the ideas. Innovation turns them into value.

The relationship is a loop. A deployed innovation brings feedback and new problems, which start a new round of creativity.



Example: Hana's time slots led to a new question: how might we help students who have no phone?

2.4.2 How they differ

	Creativity	Innovation
Focus	Producing new and useful ideas	Putting ideas into practice to create value
Typical people	Individuals and small teams	Teams, managers, and whole organizations
Resources	Time, curiosity, and imagination	Time, money, approvals, and training
Risk	Low, since ideas cost little	Higher, since implementation costs resources
Success	Originality and usefulness of the idea	Real improvement and adoption by users

2.4.3 Strengthening creativity and innovation in organizations

- **Leadership that listens.** Leaders show interest in new ideas and protect people who try them.
- **Psychological safety.** Amy Edmondson's research shows teams learn more when members feel safe to speak up and admit mistakes.
- **Time and space.** Time to think, experiment, and meet across departments.
- **Simple channels for ideas.** Suggestion systems, short idea meetings, a clear answer to every suggestion.
- **Diverse teams.** Different ages, skills, and backgrounds bring different views.
- **Learning from failure.** Review pilots that did not work and share what was learned.
- **Recognition.** Thank people for effort and results, not only for success.

2.4.4 Challenges and practical responses

Challenge	Why it happens	A possible response
Resistance to change	People fear extra work or loss of status	Involve them early and explain the benefits
Fear of failure	Mistakes are punished or ridiculed	Run small pilots and praise learning
Limited resources	Budgets and time are tight	Start with cheap prototypes and use existing tools
Short-term pressure	Daily tasks crowd out improvement	Set aside protected time each week
Rigid hierarchy	Ideas must pass through many levels	Create safe, direct channels for suggestions
Weak follow-through	No owner or plan after the pilot	Name an owner and write a rollout plan

Activity 3: two-minute project pitch

20 minutes **Use today's worksheets**

Cover these five points, in order:

1. The problem as a "How might we" question.
2. The chosen idea and why it was chosen.
3. The prototype and how it was tested.
4. The plan to deploy and scale it.
5. One challenge you expect and how you will respond.

Listeners

Give each group one positive comment and one helpful suggestion.

Chapter summary

- Creativity and innovation both follow recognizable steps that can be learned and managed.
- **Creativity:** Preparation, Incubation, Illumination, Evaluation, Implementation.
- **Innovation:** Ideation, Concept Development, Prototyping, Validation, Deployment and Scaling.
- They depend on each other, yet differ in focus, people, resources, and risk.
- Organizations strengthen them through listening leaders, psychological safety, time, diverse teams, and learning from failure.
- Common challenges are resistance, fear of failure, limited resources, and weak follow-through. Each has a practical response.

Key terms

Term	Meaning
Process	A series of steps that lead to a result
Preparation	Understanding the problem and gathering knowledge before looking for solutions
Incubation	A planned pause that lets the mind work in the background
Illumination	The moment of insight when a new idea appears
Prototype	A quick, cheap version of an idea used to learn and test
Validation	Testing an idea with real users and comparing results with goals
Scaling	Extending a successful idea to more people, offices, or places
Psychological safety	A shared belief that it is safe to speak up and admit mistakes

Review questions for discussion or homework

1. Explain the five steps of the creativity process using an example from your own workplace.
2. Why is Preparation often the most neglected step, and what are the risks of skipping it?
3. Describe the difference between Illumination in the creativity process and Ideation in the innovation process.
4. Choose one prototype idea for a problem in your office and explain how you would test it.
5. How do creativity and innovation depend on each other? Give an example of each failing without the other.
6. What are three actions a manager can take to build psychological safety in a team?

Your next steps after training

Within the next thirty days:

1. Take your project thread to your supervisor and ask permission to run your small trial.
2. Share what you learned with at least one colleague.
3. Meet again with your group after one month and compare results.

Thank you for your effort and your ideas over these two days.

Further reading

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